

Principles of Microeconomics Terminology and Key Points

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Chapter 1: Ten Principles of Economics

Ten Principles of Economics

How People Make Decisions

1. People face trade-offs
2. The cost of something is what you give up to get it
3. Rational people think at the margin
4. People respond to incentives

How People Interact

5. Trade can make everyone better off
6. Markets are usually a good way to organize economic activity
7. Governments can sometimes improve market outcomes

How the Economy as a Whole Works

8. A country's standard of living depends on its ability to produce goods and services
9. Prices rise when the government prints too much money
10. Society faces a short-run trade-off between inflation and unemployment

What is Economics?

- **Scarcity:** The limited nature of society's resources
- **Economics:** The study of how society manages its scarce resources

How People Make Decisions

- **Efficiency:** The property of society getting the most it can from its scarce resources.
- **Equality:** The property of distributing economic prosperity uniformly among the members of society
- **Opportunity cost:** Whatever must be **given up** to obtain that item
- **Rational people:** People who systematically do the best they can to achieve their objectives, given the opportunities available to them
- **Marginal change:** A small **incremental adjustment** to an existing plan of action

- **Incentive**: Something that induces a person to act, such as a *punishment or a reward*
- **Sunk cost**: Cost that is gone **regardless of what's decided next**

How People Interact

- **Market economy**: An economy that allocates resources through the **decentralized decisions** of many firms and households as they **interact in markets** for goods and services
- **Adam Smith's Invisible Hand** steers markets and households towards outcomes that maximize the well-being of society as a whole
- **Property rights** are the ability of an individual to **own and exercise control** over scarce resources
- **Market failure** is a situation in which a market left on its own **fails to allocate resources efficiently**
- **Externality**: The effect of one person's actions on a bystander (e.g. pollution)
- **Market power**: A single actor's ability to influence market prices (e.g. monopoly)

How the Economy as a Whole Works

- **Productivity** is the **quantity of goods and service** produced from **each unit** of labor input
- **Inflation** is an **increase in the overall level of prices** in the economy
- **The business cycle** refers to fluctuations in economic activity, such as employment and production

Chapter 2: Thinking Like an Economist

The Economist as a Scientist

- A **natural experiment** is a real-world event that, by chance, **exposes one group to change** while leaving a similar **comparison group** unaffected, creating something close to a controlled experiment.

The Circular-Flow Diagram

- The **Circular-Flow Diagram** is a visual model of the economy that shows how dollars flow through markets among households and firms. It **leaves out government and international trade.**

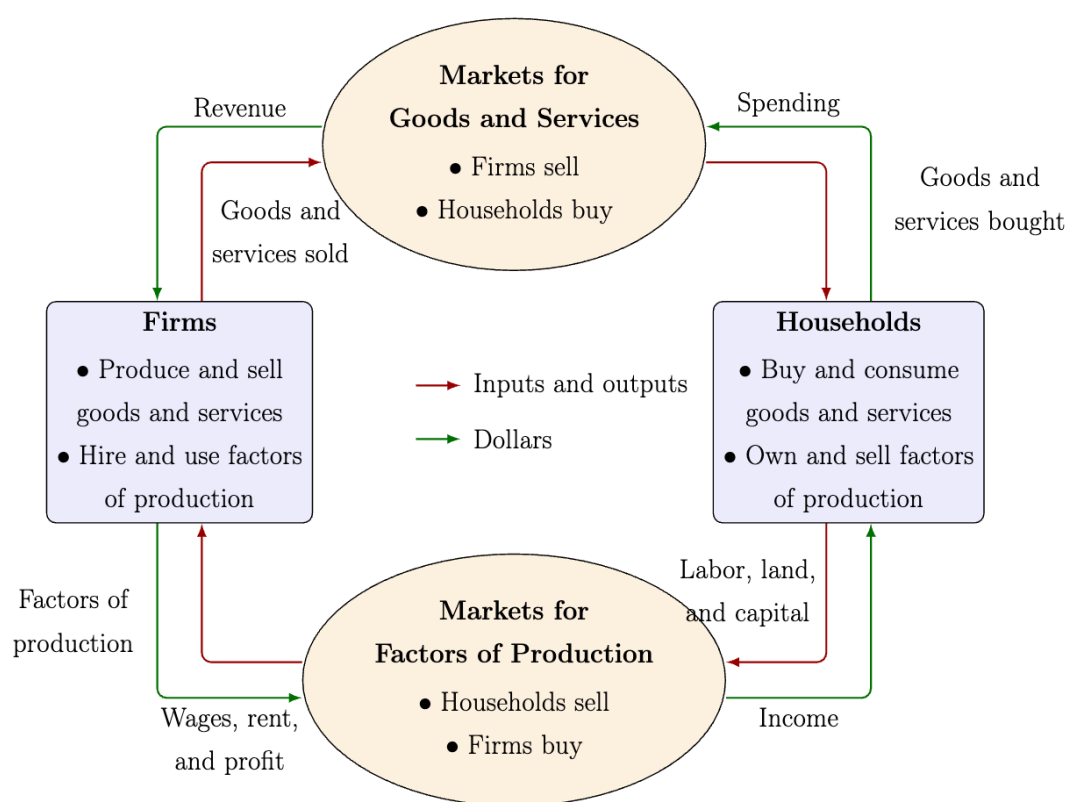


Figure 1: The circular-flow diagram

The Production Possibilities Frontier

- Production Possibilities Frontier (PPF):** A graph that shows the various combinations of output that an economy can produce, given its available factors of production and technology.
- An outcome is **efficient** if the economy is getting all it can from its scarce resources: **no more of one good can be produced** without producing less of the other.
- An outcome is **infeasible** if the current resources the economy has **are not enough** to produce the desired goods.
- An outcome is **inefficient** if the economy can **still produce more goods** with its current resources.

Microeconomics and Macroeconomics

- **Microeconomics:** The study of how **households and firms** make decisions and how they interact in **specific markets**
- **Macroeconomics:** The study of **economy-wide** phenomena, such as inflation, unemployment, and economic growth

Positive vs Normative Statements

- **Positive statements** are *descriptive*, and attempt to describe the world **as it is**. In principle, they can be confirmed or refuted with evidence
- **Normative statements** are *perscriptive*, and attempt to describe how the world **ought to be**, and cannot be settled by evidence alone.

Chapter 3: Interdependence and the Gains from Trade

- **Interdependence** refers to how we **rely on people all over the world** to provide us with goods, and in return provide something they want.
- **Absolute advantage** is the ability to produce a good using **fewer inputs** than another producer.
- **Comparative advantage** is the ability to produce a good at a **lower opportunity cost** than another producer.
- **Imports** are goods produced abroad and sold domestically
- **Exports** are goods produced domestically and sold abroad

Chapter 4: Market Forces of Supply and Demand

Markets and Competition

- A **market** is a **group of buyers and sellers** of a particular good or service.
- A **competitive market** is a market with **so many** buyers and sellers that each one has a **negligible impact** on the market price.
- A **monopoly** is a market with a **single seller**, who **sets the price** rather than taking it as given.

Demand

- The **quantity demanded** of a good is the amount that buyers are **willing and able to purchase**.
- The **law of demand** states that, other things being equal, the **quantity demanded** of a good **falls** when the **price** of the good **rises**, and falls when the price falls.
- A **demand schedule** is a **table** showing the **relationship between the price of a good and the quantity demanded**, holding everything else constant.
- A **demand curve** is a **graph** of the relationship between the price of a good and the quantity demanded.
- **Market demand** is the **sum of the individual demands** of every buyer in the market, and the market demand curve is found by adding the individual curves horizontally.

Factors that Shift a Demand Curve

- Income
- Prices of related goods
- Tastes
- Expectations
- Number of buyers

Normal and Inferior Goods

- A **normal good** is one for which **demand rises when income rises**.
- A **inferior good** is one for which **demand falls when income rises**.

Prices of Related Goods

- Two goods are **substitutes** if an **increase** in the price of one leads to an **increase** in the demand for the other.
- Two goods are **complements** if an **increase** in the price of one leads to a **decrease** in the demand for the other.

Supply

- The **quantity supplied** of a good is the amount that sellers are **willing and be able to sell** at a given price during a specified period.
- The **law of supply** states that, other things being equal, the quantity of a supplied good **rises** when the **price** of a good **rises**, and falls when the price falls.
- A **supply schedule** is a **table** showing the relationship between the *price of a good* and the *quantity supplied*, holding everything else constant.
- A **supply curve** is a **graph** of the relationship between the *price of a good* and the *quantity supplied*.
- The **market supply** is the **sum of supplies** of all sellers.

Factors that Shift the Supply Curve

- Input prices
- Technology
- Expectations
- Number of sellers

Equilibrium

- The market's **equilibrium** is the point where the **supply and demand curves intersect**. The price there is the **equilibrium price** (or **market-clearing price**) and the quantity is the **equilibrium quantity**

Surplus

- A **surplus** (or **excess supply**) is a situation in which the **quantity supplied exceeds the quantity demanded**

Shortage

- A **Shortage** (or **excess demand**) is a situation in which the **quantity demanded exceeds the quantity supplied**

Law of Supply and Demand

- The **law of supply and demand** describes the tendency of prices to adjust until **quantity supplied equals quantity demanded**

How Prices Allocate Resources

- **Price gouging** is when prices for *scarce goods* rise sharply (usually during a crisis)

Change in Supply vs Change in Quantity Supplied

- **Supply** refers to the position of the supply **curve**
- **Quantity supplied** refers to a **particular point** on the curve, i.e. the amount sellers wish to sell at a particular price

The same applies to **demand** and **quantity demanded**.

Chapter 5: Elasticity and Its Application

The Price Elasticity of Demand

- **Elasticity** is a measure of the **responsiveness** of quantity demanded or quantity supplied to a change in one of its determinants.
- The **price elasticity of demand** measures how much the **quantity demanded** of a good responds to a **change in its price**
- Demand is **elastic** if quantity demanded responds **substantially** to a price change
- Demand is **inelastic** if quantity demanded responds only **slightly** to a price change
- Demand is **unit elastic** when the elasticity is exactly 1.

Factors that Determine the Price Elasticity of Demand

- Availability of close substitutes
- Necessities vs luxuries
- Definition of the market
- Time horizon

Mathematical Formulae

Formula for Price Elasticity of Demand

$$\text{Price elasticity of demand} = \left| \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}} \right|$$

Midpoint Method for Price Elasticity of Demand

Given the initial price P_1 , the initial quantity Q_1 , the final price P_2 , and the final quantity Q_2 , we have:

$$\text{Price elasticity of demand} = \left| \frac{(Q_2 - Q_1) \div \frac{Q_2 + Q_1}{2}}{(P_2 - P_1) \div \frac{P_2 + P_1}{2}} \right|$$

In general, the “midpoint method” simply refers to using the midpoint as the denominator (even for e.g. percentage change in price or quantity).

Total Revenue

- **Total revenue** is the amount sellers receive from sales: the price received per unit times the quantity sold $P \times Q$. Without a *tax wedge*, this also equals *buyers' total spending*.

The Elasticity–Revenue Rule

Along a fixed demand curve, **elasticity** determines whether the **price effect** or the **quantity effect** on total revenue, $P \times Q$, is **stronger**.

Demand	Price rises	Price falls
Inelastic ($E < 1$)	Revenue rises	Revenue falls
Unit elastic ($E = 1$)	Revenue unchanged	Revenue unchanged
Elastic ($E > 1$)	Revenue falls	Revenue rises

Summary

With inelastic demand, price and revenue move in the same direction. With elastic demand, they move in opposite directions.

Other Demand Elasticities

- The **income elasticity of demand** measures how much the quantity demanded of a good responds to a change in **consumer's income**. It is the percentage change in quantity demanded divided by the percentage change in income.
 - Here, keep the sign. **Normal goods would be positive, inferior goods would be negative**
- The **cross-price elasticity of demand** is the percentage change in quantity demanded of **one good** divided by the percentage change in price of **another good**
 - Here, keep the sign. **Substitutes would be positive, complements would be negative**

The Price Elasticity of Supply

- The **price elasticity of supply** measures how much the **quantity supplied** of a good **responds to a change in its price**. It is the percentage in quantity supplied divided by the percentage change in price.
- Supply is **elastic** if quantity supplied responds **substantially** to price
- Supply is **inelastic** if quantity supplied responds **only slightly** to price

Applications

- Interdiction** refers to efforts to **restrict supply**, such as intercepting shipments and arresting smugglers.